A Walsh-convolution proof of two conjectures of Lachaume

Carptopus
carptopus@163.com

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Abstract

For degree-p polynomials P and Q, consider

$$\mathcal{D}_p(P,Q)(x) = \sum_{k=0}^{p} P^{(k)}(x)Q^{(p-k)}(x).$$

Lachaume conjectured that this polynomial is real-rooted when P and Q are real-rooted, and more generally that every zero of $\mathcal{D}_p(P,Q)$ lies in the convex hull of the zeros of P and Q. We observe the exact identity

$$\mathcal{D}_p(P,Q)(x) = p!\,(P \boxplus_p Q)(2x),$$

where $\boxplus_p$ is the classical Walsh symmetric additive convolution. Walsh's root-location theorem then proves both assertions in every degree. Through Lachaume's equivalences, this also establishes the corresponding semi-symmetric hyperbolicity statement. The convolution and its root-location theorem are classical; the purpose of this note is to record their direct application to Lachaume's questions.

1 The derivative sum

For complex polynomials P, Q of degree at most p, where $p \geq 1$, define

$$\mathcal{D}_p(P,Q)(x) = \sum_{k=0}^{p} P^{(k)}(x)Q^{(p-k)}(x).$$

The degree-p Walsh symmetric additive convolution, in the normalization of Marcus, Spielman and Srivastava [2], is

$$(P \boxplus_p Q)(y) = \frac{1}{p!}\sum_{k=0}^{p} P^{(k)}(y)Q^{(p-k)}(0).$$

Theorem 1. *For all complex polynomials P, Q of degree at most p,*

$$\mathcal{D}_p(P,Q)(x) = p!\,(P \boxplus_p Q)(2x).$$

Proof. Fix x and set

$$F(t) = \sum_{k=0}^{p} P^{(k)}(x+t)Q^{(p-k)}(x-t).$$

Differentiating gives

$$\begin{aligned} F'(t) = &\sum_{k=0}^{p} P^{(k+1)}(x+t)Q^{(p-k)}(x-t) \\ &- \sum_{k=0}^{p} P^{(k)}(x+t)Q^{(p-k+1)}(x-t) = 0. \end{aligned}$$

The interior terms cancel after shifting the index, while the two endpoint terms vanish because the $(p+1)$-st derivatives are zero. Hence F is constant. Evaluating at $t=0$ and $t=x$ yields

$$\mathcal{D}_p(P,Q)(x) = \sum_{k=0}^{p} P^{(k)}(2x)Q^{(p-k)}(0) = p!\,(P \boxplus_p Q)(2x).$$

□

2 Root location

The classical Walsh theorem says that if the zeros of P and Q lie in closed discs C_1 and C_2, respectively, then every zero of $P \boxplus_p Q$ lies in the Minkowski sum $C_1 + C_2$. Leake and Ryder [3, p. 1] state this complex-disc form explicitly and trace the convolution to Walsh [4].

Theorem 2. *Let $P, Q \in \mathbb{C}[X]$ have degree p. Every zero of $\mathcal{D}_p(P,Q)$ lies in the convex hull of all zeros of P and Q.*

Proof. Let S be the union of the two finite zero multisets, and let C be any closed disc containing S. Walsh's theorem puts every zero of $P \boxplus_p Q$ in $2C = C + C$. Theorem 1 therefore puts every zero of $\mathcal{D}_p(P,Q)$ in C.

The intersection of all closed discs containing a finite planar set S is its convex hull. One inclusion follows from convexity. For the other, strictly separate any point outside $\operatorname{conv}(S)$ by a line, then approximate the half-plane containing S by a sufficiently large disc whose center recedes along the opposite normal. Thus every zero of $\mathcal{D}_p(P,Q)$ belongs to $\operatorname{conv}(S)$. □

Corollary 3. *If P and Q are real-rooted, then $\mathcal{D}_p(P, Q)$ is real-rooted.*

Proof. All input zeros lie on the real line, so their convex hull is a real interval. □

3 Lachaume's conjectures

Lachaume [1, Theorem 4.9] states that his Conjecture 4 is equivalent to Corollary 3 for every degree p. Therefore Conjecture 4 holds for all $p \geq 1$. By [1, Theorem 4.8], this proves the corresponding hyperbolicity of the semi-symmetric polynomials $s_{n,p}$ for every admissible $n \geq p$.

Lachaume's Conjecture 5 is precisely Theorem 2 and hence also holds for every $p \geq 1$. The paper further records that Conjecture 4 implies Conjecture 2 in its stated nonnegative-coefficient setting. This concavity implication is not new: Lachaume's later Lovelock-gravity paper explicitly records the real-rooted diagonal criterion as an application of Walsh's theorem [5, Theorem 2.7(d)]. If Lachaume's diagonal restriction polynomial $\bar{f}_{\vec{a}}$ has actual degree $q < p$ because its leading coefficients vanish, the degree-q result gives concavity of $f^{1/q}$; composing with the increasing concave map $u \mapsto u^{q/p}$ gives concavity of $f^{1/p}$. The positive constant case is immediate. The identically zero function lies outside Lachaume's strict positive-value definition of μ-concavity; it is concave only under the usual extended nonnegative-function convention.

These implications do not address Lachaume's Conjectures 1 or 3.

4 Scope and prior work

Walsh introduced the symmetric additive convolution and proved its classical root-location property [4]. Marcus, Spielman and Srivastava [2, Definition 1.1 and Theorem 1.3] use the same convolution with a $1/p!$ normalization and record its real-rootedness preservation. Leake and Ryder [3] use the unnormalized convention and explicitly recall the Minkowski-sum statement for complex discs. Multiplication by $p!$ does not affect zeros.

Lachaume's later paper [5, Theorem 2.7(d)] already invokes Walsh's theorem to deduce concavity from real-rootedness of a diagonal restriction. Thus the broad Lachaume–Walsh connection is prior work. Theorem 1 is an elementary expression of translation covariance, not a new convolution theorem. The contribution of this note is limited to the explicit identity with its exact scaling, its application to Lachaume's Conjectures 4 and 5, and the complex convex-hull deduction. Searches completed on 4 September 2026 found no earlier source stating this identity or using it to settle both conjectures. That search result does not establish global priority, and no claim of priority for Walsh's theorem is made.

Computational check

The accompanying symbolic script verifies Theorem 1, the telescoping certificate, and the apolar coefficient identity for generic polynomials of degrees 1 through 8. It also contains a destructive control obtained by deleting one endpoint term. The script is supplementary evidence; the proof above is independent of it.

AI-assisted research disclosure

Carptopus is the responsible author and takes responsibility for the claims, proofs, source use, and released artifacts. OpenAI Codex was used for AI-assisted literature organization, proof development and stress testing, exact verification, adversarial auditing, and manuscript preparation.

References

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